

Understanding Data Visualizations

Chart Selection, Purpose & Best Practices

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Learning Objectives

- Identify the purpose of major chart types in exploratory data analysis
- Match visualization choices to data types (categorical vs. continuous)
- Recognize when a chart reveals insights versus when it misleads
- Apply best practices for clarity, accuracy, and audience communication
- Critically evaluate visualizations in research and reports

Why Visualization Matters

Key Principles

- **Human perception:** We process visuals 60,000× faster than text
- **Pattern recognition:** Charts reveal trends, outliers, clusters instantly
- **Communication:** A well-designed chart tells a story without words
- **Decision support:** Stakeholders act on clear evidence, not raw tables

Data Types Drive Chart Choice

Data Type	Examples (GSS)	Chart Families
Categorical (Nominal)	sex, marital, degree	Bar, Pie, Count
Categorical (Ordinal)	degree (ordered)	Bar, Box (with order)
Continuous (Numeric)	income, age, height	Histogram, Scatter, Line
Mixed (Cat + Num)	income by sex	Box, Violin, Point, Facet

Always start by classifying your variables. Trying to plot a categorical variable on a continuous axis (or vice versa) creates meaningless visuals. The GSS dataset contains all four types—perfect for practicing chart selection.

The Visualization Decision Tree

- ❶ **What is my question?** (Distribution? Comparison? Relationship? Composition?)
- ❷ **What are my variables?** (1 categorical? 2 continuous? Mixed?)
- ❸ **Who is my audience?** (Technical peers? Policymakers? General public?)
- ❹ **What story do I want to tell?** (Highlight inequality? Show trend? Reveal outlier?)

This four-step filter prevents "chart shopping"—picking a visually appealing but inappropriate graph. Always align chart choice with analytical intent.

Histogram: Purpose & Use

What it shows

Distribution of a single continuous variable (e.g., income, age)

- **Why use it:** Reveal shape (normal, skewed, bimodal), central tendency, spread
- **When to use:** Exploring a numeric variable; checking assumptions for statistical tests
- **GSS example:** Income distribution—right-skewed with long tail of high earners
- **Best practice:** Adjust bin count to balance detail vs. noise; add KDE line for smoothness

Histograms answer: "What values are common? Are there outliers? Is the data symmetric?" Avoid too many bins (noisy) or too few (oversimplified). For income, 30-50 bins often work well.

Histogram: Pitfalls & Alternatives

- **Pitfall:** Bin boundaries can artificially create or hide patterns
- **Pitfall:** Hard to compare distributions across groups (use faceting instead)
- **Alternative:** Kernel Density Estimate (KDE) for smooth distribution curves
- **Alternative:** Box plot when you care more about quartiles than full shape

Rule: Use histograms for initial exploration; switch to KDE or box plots for comparison. Always try multiple bin widths when exploring. If conclusions change dramatically with bin choice, the pattern may not be robust. KDE avoids binning artifacts but can oversmooth small datasets.

Box Plot: Purpose & Use

What it shows

Five-number summary (min, Q1, median, Q3, max) + outliers for a continuous variable

- **Why use it:** Compact summary of center, spread, skew, and extreme values
- **When to use:** Comparing distributions across categories; detecting outliers
- **GSS example:** Income by marital status—compare medians and IQRs across groups
- **Best practice:** Show mean as a distinct marker; label outliers if contextually important

Box plots excel at comparison. The interquartile range ($IQR = Q3 - Q1$) shows where the middle 50% of data lies. Outliers (points beyond $1.5 \times IQR$) warrant investigation—are they errors or genuine extremes?

Box Plot: Interpretation Guide

Visual Feature	What it tells you
Median line position	Central tendency; skew if off-center
Box height (IQR)	Variability; taller = more spread
Whisker length	Range of typical values
Points beyond whiskers	Potential outliers
Notches (if shown)	Confidence interval for median

If the median is closer to Q1, the distribution is right-skewed (common for income). If whiskers are asymmetric, the tails differ. Always report sample sizes—small groups can produce misleading box plots.

Violin Plot: Purpose & Use

What it shows

Combination of box plot + kernel density estimate—full distribution shape by category

- **Why use it:** Reveal multimodality, skew, and density that box plots hide
- **When to use:** Comparing distributions where shape matters (not just center/spread)
- **GSS example:** Age distribution by education level—do graduate-degree holders cluster at certain ages?
- **Best practice:** Use inner markers (box or quartiles) to retain summary stats; avoid overcrowding

Violin plots answer: "Is this group's distribution unimodal or bimodal? Are there subgroups?" For GSS, income by degree might show bimodality within "Bachelor" (early-career vs. established professionals).

Violin Plot: When NOT to Use

- **Avoid when:** Sample size per group < 20 (KDE becomes unreliable)
- **Avoid when:** Audience is non-technical (violin shapes can confuse)
- **Avoid when:** You only need to compare medians (use box plot instead)
- **Alternative:** Ridgeline (joy) plots for many groups; strip plots for small N

Rule: Violin plots trade simplicity for detail—use only when distribution shape is analytically critical. For policy audiences, a simple box plot with clear labels often communicates more effectively. Reserve violins for exploratory analysis or technical reports where nuance matters.

Count Plot (Bar Chart): Purpose & Use

What it shows

Frequency or proportion of categories for a categorical variable

- **Why use it:** Show composition, dominance, or rarity of categories
- **When to use:** Summarizing survey responses, demographic breakdowns
- **GSS example:** Distribution of education levels—what
- **Best practice:** Order bars by frequency or logical sequence; add value labels

Pie Chart: Purpose & Strong Caveats

What it shows

Proportional composition of a whole (categories sum to 100%)

- **Why use it (rarely):** Emphasize part-to-whole relationships for 5 categories
- **When to avoid:** Comparing similar-sized slices; ≥ 5 categories; precise value reading
- **GSS example:** Marital status proportions—but bar chart is clearer for comparison
- **Best practice:** Order slices largest→smallest clockwise from top; label with %

Research shows humans judge angles (pie slices) less accurately than lengths (bar heights). Use pie charts only for high-level storytelling to non-technical audiences, never for analytical comparison.

KDE Plot (Density Curve): Purpose & Use

What it shows

Smoothed probability density of a continuous variable

- **Why use it:** Reveal distribution shape without binning artifacts; compare groups via overlay
- **When to use:** Checking normality; visualizing theoretical distributions; overlaying groups
- **GSS example:** Overlay income density for males vs. females to compare earning distributions
- **Best practice:** Adjust bandwidth parameter; use transparency for overlapping curves

KDE plots answer: "Is this distribution unimodal or multimodal? How skewed is it?" For income, KDE clearly shows the right tail. Bandwidth controls smoothness—too small = noisy, too large = oversmoothed.

Scatter Plot: Purpose & Use

What it shows

Relationship between two continuous variables (each point = one observation)

- **Why use it:** Detect correlation, clusters, outliers, non-linear patterns
- **When to use:** Exploring associations; checking linearity before regression
- **GSS example:** Age vs. income—does earnings increase with experience?
- **Best practice:** Add transparency (alpha) for overplotting; include trend line if helpful

Scatter plots are foundational for correlation analysis. Look for: direction (positive/negative), strength (tightness), form (linear/curved), and outliers. For GSS, age-income may show a curve (rising then plateauing).

Scatter Plot: Enhancements for Clarity

- **Add marginal histograms:** Show univariate distributions alongside bivariate relationship
- **Color by category:** Reveal group-specific patterns (e.g., income-age by education)
- **Add regression line:** Quantify linear trend; report R^2 for explained variance
- **Use hexbin for large N:** Replace overlapping points with density-colored hexagons

Rule: Every enhancement should answer a specific analytical question—avoid decorative clutter. For GSS with 2,500+ points, $\alpha=0.3$ or hexbin prevents overplotting. Coloring by degree might reveal that the age-income slope differs by education level—a key interaction effect.

Line Plot: Purpose & Use

What it shows

Trend of a continuous variable over an ordered dimension (time, age, sequence)

- **Why use it:** Highlight trajectories, cycles, or monotonic changes
- **When to use:** Time series data; age cohorts; ordered categories
- **GSS example:** Mean income by age group—lifecycle earnings profile
- **Best practice:** Connect only meaningful sequences; avoid implying continuity where none exists

Line plots imply continuity and order. Never use for nominal categories (e.g., connecting "Divorced" to "Married" makes no sense). For age cohorts in GSS, lines show how earnings evolve across the lifespan.

Line Plot: Confidence Intervals

- **Why add CI:** Show uncertainty around estimates (e.g., mean income per age group)
- **Interpretation:** "We are 95% confident the true mean lies within this band"
- **GSS application:** Wide CIs at older ages may reflect smaller sample sizes
- **Best practice:** Use shaded bands (not error bars) for cleaner visuals; label confidence level

Confidence intervals prevent overinterpreting small differences. If CIs overlap substantially between age groups, the income difference may not be statistically significant. Always consider sample size when evaluating precision.

Grouped Box Plot: Purpose & Use

What it shows

Distribution of a continuous variable across levels of a categorical variable

- **Why use it:** Compare central tendency, spread, and outliers across groups
- **When to use:** Testing group differences; exploring effect modifiers
- **GSS example:** Income distribution by education level—does higher degree = higher earnings?
- **Best practice:** Order categories logically (e.g., education ascending); add sample sizes

Grouped box plots answer: "Do groups differ in typical value? In variability? In outlier prevalence?" For GSS, we expect median income to rise with education, but check if variability also increases (e.g., graduate degrees may have wider income ranges).

Grouped Box Plot: Statistical Context

- **Not a hypothesis test:** Visual differences $\neq$ statistical significance
- **Follow up with:** ANOVA (for ≥ 2 groups) or t-test (for 2 groups) on the same data
- **Check assumptions:** Normality and equal variance affect test validity
- **Report effect size:** Not just p-values—how large is the difference practically?

Rule: Visualizations generate hypotheses; statistical tests evaluate them. A box plot might show males have higher median income than females in GSS. But is this difference larger than expected by chance? That requires a t-test. Always pair visuals with appropriate inference.

Point Plot with Error Bars: Purpose & Use

What it shows

Estimated mean (or median) for each category, with confidence intervals

- **Why use it:** Emphasize point estimates and uncertainty; cleaner than box plots for many groups
- **When to use:** Reporting model coefficients; comparing adjusted means
- **GSS example:** Average income by marital status, controlling for age
- **Best practice:** Use consistent markers; label confidence level; avoid overlapping CIs

Point plots focus attention on the estimate, not the full distribution. Ideal when the research question is "What is the average effect?" rather than "What is the full range of outcomes?" For policy, point estimates with CIs are often most actionable

Heatmap (Correlation Matrix): Purpose & Use

What it shows

Pairwise correlation coefficients between multiple continuous variables

- **Why use it:** Quickly scan for strong relationships; detect multicollinearity
- **When to use:** Pre-modeling diagnostics; exploratory correlation screening
- **GSS example:** Correlations among age, income, height, weight, hours worked
- **Best practice:** Use diverging color scale (red/blue); annotate with values; sort variables logically

Heatmaps answer: "Which variables move together?" For GSS, we expect height-weight correlation 0.7, but income-age may be weaker. Remember: correlation $\neq$ causation, and non-linear relationships won't appear.

Heatmap: Interpretation Guidelines

Correlation (r)	Interpretation
$-0.3 \leq r \leq 0.3$	Weak or no linear relationship
$0.3 \leq r \leq 0.5$	Moderate linear relationship
$ r \geq 0.5$	Strong linear relationship
$r \geq 0$	Positive association (both increase)
$r \leq 0$	Negative association (one increases, other decreases)

Always report the correlation coefficient alongside the heatmap. A "strong" visual color may correspond to $r=0.4$ —statistically significant but practically modest. Context matters: in social sciences, $r=0.3$ can be meaningful.

Joint Plot: Purpose & Use

What it shows

Bivariate scatter plot with marginal univariate distributions (histograms or KDE)

- **Why use it:** Show relationship + individual distributions in one cohesive view
- **When to use:** Initial exploration of two continuous variables
- **GSS example:** Height vs. weight—scatter shows correlation, margins show each variable's spread
- **Best practice:** Use consistent binning/KDE bandwidth; add correlation coefficient in title

Joint plots efficiently answer three questions: (1) Are X and Y related? (2) What's the distribution of X? (3) What's the distribution of Y? For GSS, this quickly confirms the expected positive height-weight correlation.

Faceted Plots (Small Multiples): Purpose & Use

What it shows

Same chart type repeated across subsets of data (e.g., one histogram per education level)

- **Why use it:** Compare distributions or relationships across categories without visual clutter
- **When to use:** When hue/color would overcrowd a single plot; for detailed group comparison
- **GSS example:** Income histogram faceted by gender—direct visual comparison of earning distributions
- **Best practice:** Use consistent axis scales across facets; label each facet clearly

Faceting follows the "small multiples" principle: many similar charts enable rapid comparison. For GSS, faceting income by degree reveals whether the right-skew is consistent across education levels or unique to certain groups.

Faceted Plots: Design Principles

- **Align axes:** Same scale enables direct comparison; different scales hide patterns
- **Order facets logically:** By frequency, alphabetically, or by a meaningful metric
- **Limit number:** ≤ 6 facets can overwhelm; consider grouping or filtering first
- **Add reference lines:** A vertical line at median income helps compare across facets

Rule: Facets should reduce cognitive load, not increase it. For GSS, faceting by marital status (5 categories) works well. Faceting by both marital status AND gender (10 combinations) may require two separate facet grids or a different approach like grouped box plots.

Pair Plot (Scatterplot Matrix): Purpose & Use

What it shows

All pairwise scatter plots for a set of continuous variables, with distributions on diagonal

- **Why use it:** Comprehensive view of multivariate relationships in one figure
- **When to use:** Initial EDA of multiple numeric variables; checking for multicollinearity
- **GSS example:** Pair plot of age, income, height, weight, hours worked
- **Best practice:** Limit to 5 variables; use corner parameter to avoid redundancy; add correlation values

Pair plots answer: "What are all the bivariate relationships in my dataset?" For GSS, this quickly reveals expected correlations (height-weight) and surprises (maybe hours worked and income are weakly related). Use sparingly—overuse leads to visual fatigue.

3D Scatter Plot: Purpose & Strong Caveats

What it shows

Relationship among three continuous variables in three-dimensional space

- **Why use it (rarely):** Explore complex interactions when 2D plots are insufficient
- **When to avoid:** Static reports; non-technical audiences; when rotation isn't possible
- **GSS example:** Age $\times$ hours worked $\times$ income, colored by BMI
- **Best practice:** Use interactive tools (Plotly); provide 2D projections as backup; never use for final publication

3D plots often obscure more than they reveal in static images due to occlusion and perspective distortion. Reserve for interactive exploration. For GSS, a faceted 2D approach (e.g., income vs. age, faceted by hours-worked bins) is usually clearer.

Ridgeline Plot (Joy Plot): Purpose & Use

What it shows

Overlapping density curves for a continuous variable across multiple categories

- **Why use it:** Elegant comparison of distribution shapes across many groups
- **When to use:** When distribution shape (not just center) is the focus; for ordinal categories
- **GSS example:** Income density across education levels—do curves shift right with higher education?
- **Best practice:** Order categories meaningfully; use transparency; avoid ≥ 8 groups

Ridgeline plots excel at showing how an entire distribution evolves across a sequence. For GSS education levels (ordered), we'd expect income densities to shift rightward and potentially widen with higher education. The overlapping design saves space vs. separate KDE plots.

Stacked Bar Chart: Purpose & Use

What it shows

Composition of categories within groups, where bars sum to 100% (or total count)

- **Why use it:** Show part-to-whole relationships across multiple groups
- **When to use:** Comparing proportional composition (e.g., marital status by gender)
- **GSS example:** Marital status distribution within each gender—do proportions differ?
- **Best practice:** Limit to 4 component categories; order segments consistently; add value labels

Stacked bars answer: "How does the makeup of Group A differ from Group B?" For GSS, if "Married" is a larger slice for males than females, that's a demographic insight. But avoid stacking too many categories—comparison becomes difficult.

Stacked Bar Chart: Limitations

- **Hard to compare non-base segments:** Only the bottom segment aligns to a common baseline
- **Alternative:** Grouped (side-by-side) bar charts for precise comparison of all categories
- **Alternative:** Mosaic plots for formal contingency table visualization
- **Rule:** Use stacked bars for part-to-whole emphasis; use grouped bars for category comparison

For GSS: If comparing "Divorced" proportion between genders, a grouped bar chart is clearer than stacked. Cognitive research shows humans compare lengths aligned to a common baseline most accurately. Stacked bars violate this for all but the bottom segment. Reserve stacking for when the total (bar height) is also meaningful.

Mosaic Plot: Purpose & Use

What it shows

Proportional areas representing cell frequencies in a contingency table (two categorical variables)

- **Why use it:** Visualize association between two categorical variables; area frequency
- **When to use:** Exploring relationships like education $\times$ marital status
- **GSS example:** Do education levels have different marital status distributions?
- **Best practice:** Add residual shading to highlight cells that deviate from independence

Mosaic plots answer: "Is there an association between these two categorical variables?" For GSS, if "Graduate" degree holders have a larger "Married" tile than expected under independence, that suggests an association. Residual shading (blue/red) makes deviations visually obvious.

Lorenz Curve & Gini Coefficient: Purpose & Use

What it shows

Cumulative share of income vs. cumulative share of population; measures inequality

- **Why use it:** Quantify and visualize economic inequality
- **When to use:** Policy analysis; cross-group or cross-time inequality comparison
- **GSS example:** Income inequality within the survey sample; compare by gender or education
- **Best practice:** Plot the line of perfect equality (45°); report Gini coefficient (0=equality, 1=inequality)

Lorenz curves answer: "How unequally is income distributed?" The further the curve bows below the equality line, the greater the inequality. For GSS, we'd expect a noticeable bow given income's right-skew. Gini allows numerical comparison across subgroups.

Q-Q Plot (Quantile-Quantile): Purpose & Use

What it shows

Sample quantiles vs. theoretical quantiles (usually normal distribution)

- **Why use it:** Assess if a variable follows a theoretical distribution (e.g., normality)
- **When to use:** Checking assumptions for parametric tests (t-test, ANOVA, regression)
- **GSS example:** Is income normally distributed? (Spoiler: almost never)
- **Best practice:** Add reference line; focus on middle quantiles; note tail deviations

Q-Q plots answer: "Does my data match the distribution I'm assuming?" For GSS income, points will curve above the line at high quantiles—confirming right-skew. This justifies using non-parametric tests or transforming income (e.g., log) for modeling.

Radar Chart (Spider Plot): Purpose & Strong Caveats

What it shows

Multivariate profile of a group across multiple normalized variables on radial axes

- **Why use it (rarely):** Compare holistic profiles across groups (e.g., education levels)
- **When to avoid:** Precise value reading; ≥ 6 variables; non-technical audiences
- **GSS example:** Normalized profiles of age, income, hours worked by education level
- **Best practice:** Normalize variables to 0-1 scale; order axes logically; use transparency for overlap

Radar charts are visually striking but cognitively demanding. Humans struggle to compare non-adjacent axes. For GSS, a table of means or small multiples of bar charts is usually clearer. Reserve radar charts for dashboards where pattern recognition $\geq$ precise comparison.

Parallel Coordinates Plot: Purpose & Use

What it shows

Multivariate observations as lines crossing parallel axes (one axis per variable)

- **Why use it:** Explore patterns across many continuous variables simultaneously
- **When to use:** High-dimensional EDA; identifying clusters or outliers in multivariate space
- **GSS example:** Lines for each respondent across age, income, height, weight, hours worked
- **Best practice:** Normalize axes; reorder axes to reveal patterns; use brushing for interaction

Parallel coordinates answer: "Are there subgroups with similar multivariate profiles?" For GSS, we might see a cluster of high-income, older, high-hours respondents. However, with 2,500+ lines, overplotting is severe—use sampling or aggregation first.

Bubble Chart: Purpose & Use

What it shows

Scatter plot where point size encodes a third continuous variable

- **Why use it:** Add a third dimension to bivariate relationships without facets
- **When to use:** When the third variable is secondary but still informative
- **GSS example:** Age vs. income scatter, with bubble size = hours worked
- **Best practice:** Use area (not radius) for size encoding; add legend; avoid extreme size ranges

Bubble charts answer: "Does the third variable modify the X-Y relationship?" For GSS, if larger bubbles (more hours) cluster in high-income regions, that suggests a three-way relationship. But size perception is imprecise—use only for approximate comparisons.

Error Bar Charts: Purpose & Interpretation

What it shows

Point estimate (mean/median) with vertical lines indicating uncertainty (CI, SD, SE)

- **Why use it:** Communicate precision of estimates; enable visual significance assessment
- **When to use:** Reporting group comparisons; model results; meta-analyses
- **GSS example:** Mean income by education level with 95% confidence intervals
- **Best practice:** Label what the bars represent (CI vs. SD); avoid overlapping bars; use consistent markers

Error bars answer: "How confident are we in this estimate?" For GSS, if CIs for "Bachelor" and "Graduate" income don't overlap, that suggests a statistically significant difference. But remember: non-overlapping 95% CIs $p < 0.01$, not $p < 0.05$ —be precise in interpretation.

Choosing Between Similar Charts: Decision Guide

Question	Better Choice	Why
Compare distributions?	Box plot	Focus on quartiles, outliers
Show full distribution shape?	Violin/KDE	Reveal multimodality, skew
Compare many groups?	Faceted plots	Avoid color overload
Show part-to-whole?	Stacked bar (4 cats)	Emphasize composition
Compare category proportions?	Grouped bar	Align baselines for accuracy
Explore 2 continuous vars?	Scatter plot	Show individual observations
Add third continuous var?	Bubble or facet	Context-dependent

This table is a quick-reference heuristic. Always return to your analytical question: "What do I need to learn from this data?" The chart is a means to that end, not the end itself.

Common Visualization Pitfalls

- **Misleading axes:** Truncated y-axis exaggerates differences
- **Overplotting:** Too many points obscure patterns (use alpha, hexbin, or sampling)
- **Color misuse:** Non-colorblind-friendly palettes; using color for ordinal data
- **Chart junk:** Unnecessary 3D effects, decorations, or gridlines
- **Ignoring uncertainty:** Reporting point estimates without confidence intervals
- **Wrong chart type:** Using pie charts for comparison; line plots for nominal categories

Avoid these pitfalls by adhering to principles: maximize data-ink ratio, ensure accessibility, and prioritize clarity over creativity. For GSS reporting, simplicity and accuracy trump visual flair.

Accessibility in Visualization

- **Colorblindness:** Use palettes like viridis, plasma, or ColorBrewer's "Set2"; avoid red-green
- **Print friendliness:** Ensure charts are interpretable in grayscale
- **Text size:** Labels and legends must be readable when projected or printed
- **Alternative text:** Provide descriptions for screen readers in digital reports
- **Cognitive load:** Limit to one main message per chart; use annotations to guide attention

Inclusive design ensures your insights reach all audiences. For GSS policy briefs, accessibility isn't optional—it's ethical. Test charts with colleagues who have visual impairments when possible.

Storytelling with Visualizations

- 1 **Start with the question:** What decision or insight does this support?
- 2 **Choose the simplest adequate chart:** Don't use a violin plot if a box plot suffices
- 3 **Annotate strategically:** Highlight key values, trends, or outliers with text/arrows
- 4 **Sequence logically:** Build understanding step-by-step in multi-chart narratives
- 5 **End with action:** What should the viewer do with this insight?

Great visualization is storytelling with evidence. For GSS, a narrative might be: "Income inequality exists (Lorenz curve), varies by education (faceted box plots), and persists after controlling for age (point plots with CIs)—suggesting education policy could reduce disparities."

Validating Your Visualization

- **Peer review:** Ask a colleague: "What's the main takeaway from this chart?"
- **Reverse-engineer:** Could someone recreate the key numbers from the visual alone?
- **Stress test:** Does the chart still make sense with different data subsets?
- **Audience test:** Show to a representative viewer—do they interpret it as intended?
- **Ethical check:** Does this visualization respect participant privacy and avoid stigmatization?

Validation prevents miscommunication. For GSS, ensure that visualizing income by demographic groups doesn't reinforce stereotypes. Always contextualize findings: correlation causation, sample population.

Summary: The Visualization Mindset

Core Principles

- **Purpose-driven:** Chart choice follows analytical question
- **Data-type aware:** Match visualization to variable measurement level
- **Audience-centered:** Design for the viewer's knowledge and needs
- **Ethically grounded:** Prioritize truth, clarity, and inclusivity
- **Iterative:** Refine based on feedback and new insights

Mastering visualization isn't about memorizing chart types—it's about cultivating a mindset of intentional, ethical communication. With the GSS dataset, this mindset transforms raw numbers into actionable insights about education, income, and well-being.

Thank You!

Questions & Discussion

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"The goal is to turn data into information, and information into insight."

— Carly Fiorina

Thank you for engaging with these visualization principles. I encourage you to apply this framework to your own data projects. Remember: the best chart is the one that helps your audience understand, decide, and act. I'm available for